

ISLE OF SHOALS, NH, USA

WMO: 994270

Lat: 42.9672N

Lon: 70.6233W

Elev: 43

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	11.6	-9.2	3.4	10.4	-5.4	4.1	14.4	47.7	30.0	42.8	23.7	25.2	300	0.798

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	81.1	68.5	78.0	66.9	75.3	66.1	72.0	76.6	70.4	74.6	69.0	72.8	15.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.6	113.0	73.8	69.0	106.8	72.1	67.8	102.1	71.1	35.7	76.5	34.3	75.0	33.2	72.8	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
41.3	35.8	32.2	1.2	88.3	5.0	3.2	-2.4	90.7	-5.3	92.5	-8.1	94.3	-11.7	96.7		
			0.1	74.8	4.6	2.2	-3.2	76.4	-5.8	77.7	-8.4	79.0	-11.7	80.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	28.4	30.0	35.5	44.7	53.0	61.9	68.2	67.5	62.1	52.9	43.4	35.4
	DBStd	15.34	9.81	8.36	7.39	6.30	6.31	6.09	4.61	4.10	5.10	5.74	7.34	7.77
	HDD50	2611	670	561	453	183	36	1	0	0	0	35	218	454
	HDD65	6244	1135	981	914	610	379	131	18	19	116	377	647	917
	CDD50	2130	1	0	4	23	128	357	565	543	363	124	21	1
	CDD65	290	0	0	0	0	6	38	118	97	29	2	0	0
	CDH74	869	0	0	1	3	26	150	394	220	70	5	0	0
	CDH80	134	0	0	0	0	4	24	68	29	9	0	0	0
Wind	WSAvg	16.4	19.9	19.1	18.2	16.7	15.3	14.1	13.4	12.7	13.6	16.5	18.0	19.4
Precipitation	PrecAvg	47.6	3.9	3.5	4.4	4.1	3.6	3.9	3.3	3.5	4.0	4.5	4.0	4.5
	PrecMax	65.7	6.5	8.5	16.6	10.1	16.8	11.3	8.5	8.0	8.8	13.3	8.0	7.3
	PrecMin	33.9	1.1	1.6	0.7	0.4	0.6	0.5	1.2	1.0	1.6	0.2	1.1	1.5
	PrecStd	7.6	1.2	1.5	3.2	2.2	3.0	2.5	1.9	1.9	2.0	3.2	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	51.9	65.5	70.9	77.3	83.5	85.5	83.9	80.2	73.2	64.2	56.2
		MCWB	51.3	44.6	52.8	55.7	61.0	68.9	72.1	69.0	68.6	61.1	57.5	52.7
	2%	DB	48.4	46.8	53.2	61.7	71.5	78.4	81.7	79.1	75.5	66.3	58.8	51.3
		MCWB	45.6	41.5	44.4	51.2	60.0	65.0	70.2	68.1	65.6	59.5	52.8	48.1
	5%	DB	45.1	43.1	47.8	56.9	66.2	74.3	78.3	76.3	72.1	62.6	55.6	48.9
		MCWB	41.7	38.6	41.7	48.8	58.6	64.5	68.0	67.1	65.2	58.2	51.0	45.9
	10%	DB	41.4	40.5	44.2	53.0	61.9	71.0	75.3	73.7	69.2	60.1	53.2	46.2
		MCWB	38.2	36.8	39.6	45.6	55.7	62.9	66.6	66.4	64.0	56.2	49.6	43.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	47.6	55.1	58.8	66.5	71.9	75.4	72.9	71.3	64.8	59.2	53.8
		MCDB	53.6	49.7	61.3	65.3	72.6	77.0	81.8	77.3	75.6	67.4	62.4	55.4
	2%	WB	46.7	42.3	46.8	54.0	63.2	69.0	73.0	71.1	69.3	62.0	55.0	49.8
		MCDB	47.6	44.3	50.7	59.9	67.7	75.0	77.9	75.1	73.1	64.6	56.6	51.3
	5%	WB	42.4	39.7	42.8	50.4	60.2	66.5	71.0	69.8	67.4	59.8	52.6	47.1
		MCDB	44.2	41.9	45.7	55.2	64.4	70.9	75.4	73.5	70.8	61.6	54.5	48.4
	10%	WB	38.3	37.0	40.6	47.3	57.3	64.4	69.2	68.6	65.3	57.7	50.5	43.9
		MCDB	41.0	39.7	43.5	51.3	60.8	68.3	72.9	72.2	68.3	59.9	53.1	45.9
Mean Daily Temperature Range	5% DB	MDBR	11.3	10.8	9.8	9.5	9.7	10.0	9.7	9.1	8.7	8.5	9.0	10.0
		MCDBR	13.7	12.8	14.7	16.7	17.9	16.8	14.8	13.1	13.1	11.7	11.3	12.0
		MCWBR	14.2	11.3	10.6	10.9	10.8	8.8	7.3	7.0	7.4	8.7	10.2	11.9
	5% WB	MCDBR	13.5	11.6	12.7	14.8	14.9	14.1	11.4	10.1	10.7	9.7	10.0	12.0
		MCWBR	14.8	11.9	11.1	11.9	11.4	9.4	7.8	7.1	8.0	8.9	10.3	13.2
Clear-Sky Solar Irradiance	taub	0.291	0.300	0.321	0.360	0.396	0.445	0.461	0.428	0.376	0.334	0.312	0.295	
	taud	2.508	2.504	2.502	2.427	2.366	2.265	2.225	2.305	2.433	2.546	2.565	2.564	
	Ebn at Noon	272	288	294	288	279	264	258	263	271	272	262	261	
	Edh at Noon	22	26	30	35	38	42	43	39	32	25	21	20	
All-Sky Solar Radiation	RadAvg	538	815	1150	1469	1645	1768	1864	1686	1350	885	587	438	
	RadStd	36	66	75	158	181	154	121	75	82	77	49	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	-250	+230	+77
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	+61

Nomenclature: See separate page